



DATA SCIENCE &
ARTIFICIAL INTELLIGENCE

Probabilistic Machine Learning

Kernel methods and Gaussian Processes

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Dual formulation of linear regression

$$y = f(x, w) + \epsilon \quad \epsilon \sim \mathcal{N}(0, \beta^{-1}) \quad f(x, w) = w_0 \phi_0(x) + \dots + w_{M-1} \phi_{M-1}(x)$$

$$w_M = \left(\Phi^T \Phi \right)^{-1} \Phi^T \underline{y} \quad \Phi = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & \cdots & \phi_{M-1}(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{pmatrix}.$$

w_M belongs to $\langle \phi(x_1), \dots, \phi(x_n) \rangle$

$$w = \sum_{j=1}^N a_j \phi(x_j)$$

$$K = (k_{ij}) \quad k_{ij} = k(x_i, x_j) \quad K^i i^{th} \text{ column} \quad w^T \phi(x_i) = a^T K^i$$

Change variables from w (primal) to a (dual)

Regularised loss becomes $\sum_{n=1}^N (y_n - a^T K^n)^2 + \lambda a^T K a$

Dual formulation of linear regression

$$\hat{a} = (k + \lambda I)^{-1} \underline{y} \quad \text{ML solution}$$

prediction at x^*

$$y(x^*) = K_*^T (K + \lambda I)^{-1} \underline{y}$$
$$K_* = (k(x_*, x_1), \dots, k(x_*, x_n))$$

This is the **Kernel Trick**: we just need to know how to compute the **kernel function** $k(x_1, x_2)$

k is the scalar product in feature space

Cost of regression: from $O(M^3)$ to $O(N^3)$

Kernel functions

kernels can be seen as linear operators on functions.

their eigenfunctions $\int k(\mathbf{x}, \mathbf{y}) \phi(\mathbf{x}) d\mu(\mathbf{x}) = \lambda \phi(\mathbf{y})$

span a possibly infinite dimensional function space.

Gaussian kernel

$$k(\mathbf{x}, \mathbf{y}) = \alpha \exp \left[- \frac{\|\mathbf{x} - \mathbf{y}\|^2}{\lambda^2} \right]$$

Automatic Relevance detection
Gaussian kernel

$$k(\mathbf{x}, \mathbf{y}) = \alpha \exp \left[- \sum_i \frac{|x_i - y_i|^2}{\lambda_i^2} \right]$$

Rational quadratic kernel

$$k_{QR}(r) = \left(1 + \frac{r^2}{2\alpha\ell^2} \right)^{-\alpha}$$

More complex kernels can be obtained by sum, product, and so on.

Random functions

Definition 12.3.1 *A random function (or stochastic process) is an infinite collection of random variables, indexed by the argument of the function.*

A stochastic process on a (regular) space $\mathcal{F}$ is characterised by its finite dimensional projections

$$p(f(x_1), \dots, f(x_n)) \text{ for each } n, x_1, \dots, x_n$$

$$\mathcal{F} = \{f(\mathbf{x}) = w_0\phi_0(\mathbf{x}) + \dots + w_{M-1}\phi_{M-1}(\mathbf{x})\}$$

M-dim vector space with basis $\Phi(\mathbf{x}) = (\phi_j(\mathbf{x}))$

A distribution on $\mathcal{F}$ is a distribution on parameters w

If $\mathbf{w} \sim \mathcal{N}(0, \mathbb{I})$ then $f(\mathbf{x}) \sim \mathcal{N}(0, \phi^T \phi)$

$$p(f(\mathbf{x}_1), f(\mathbf{x}_2)) = \mathcal{N}\left(0, \begin{pmatrix} \Phi(\mathbf{x}_1) \\ \Phi(\mathbf{x}_2) \end{pmatrix}^T \mathbb{I} \begin{pmatrix} \Phi(\mathbf{x}_1) \\ \Phi(\mathbf{x}_2) \end{pmatrix}\right)$$

$$\text{cov}(f(\mathbf{x}_1), f(\mathbf{x}_2)) = \begin{pmatrix} \Phi^T(\mathbf{x}_1)\Phi(\mathbf{x}_1) & \Phi^T(\mathbf{x}_1)\Phi(\mathbf{x}_2) \\ \Phi^T(\mathbf{x}_2)\Phi(\mathbf{x}_1) & \Phi^T(\mathbf{x}_2)\Phi(\mathbf{x}_2) \end{pmatrix} := \begin{pmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & k(\mathbf{x}_1, \mathbf{x}_2) \\ k(\mathbf{x}_1, \mathbf{x}_2) & k(\mathbf{x}_2, \mathbf{x}_2) \end{pmatrix}$$

Gaussian Processes

Definition 12.4.1 *A Gaussian Process (GP) is a stochastic process indexed by a continuous variable $\mathbf{x} \in \mathbb{R}^n$, such that all finite dimensional distributions are multivariate Gaussian.*

$f \sim GP(\mu, K)$ if and only if

$$\forall \mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^n, p(\mathbf{f}) = p(f(\mathbf{x}_1), \dots, f(\mathbf{x}_N)) = \mathcal{N}(\underline{\mu}, K)$$

mean $\underline{\mu} = (\mu(\mathbf{x}_1), \dots, \mu(\mathbf{x}_n))$

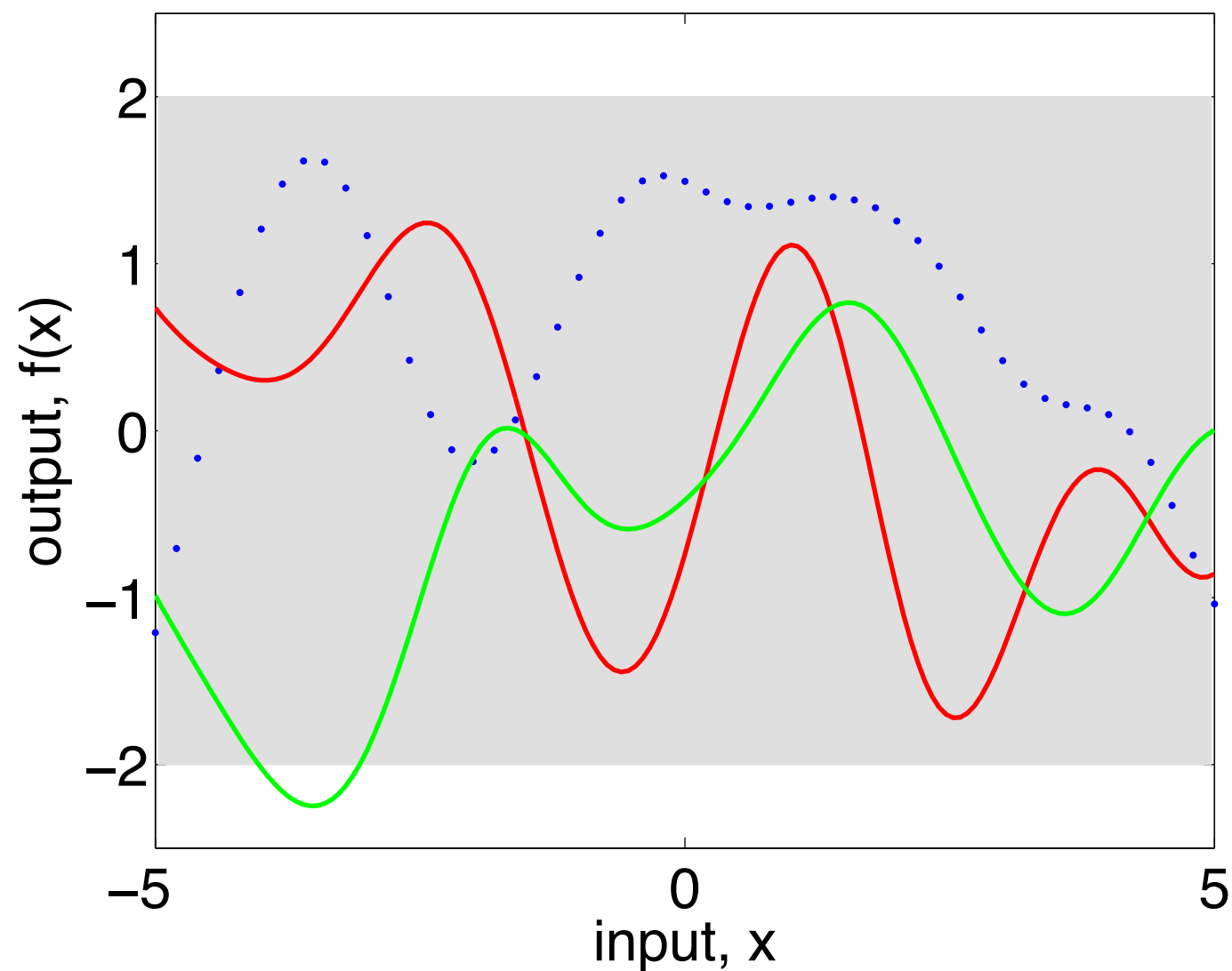
covariance matrix $K = (K_{ij})$ (with $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$)

K has to be symmetric and positive definite $\forall \mathbf{x}_1, \dots, \mathbf{x}_N$.

Example

$$\mu(x) \equiv 0 \quad k(x, x') = \exp\left[-\frac{1}{2}|x - x'|^2\right]$$

$$\mathbf{f}^* \sim \mathcal{N}(\mathbf{0}, K(X^*, X^*))$$



GP regression

$$\mathbf{y}(\mathbf{x}) = \mathbf{f}(\mathbf{x}) + \varepsilon \text{ with } \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

$$\mathbf{x} = x_1, \dots, x_N$$

observations

$$\mathbf{x}^\star = x_1^\star, \dots, x_\nu^\star,$$

predictions

$$f \sim GP(\mu, K) \quad cov(\mathbf{y}) = K(\mathbf{x}, \mathbf{x}) + \sigma^2 \mathbb{I}.$$

$$p \begin{pmatrix} \mathbf{y} \\ \mathbf{f}^\star \end{pmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} K(\mathbf{x}, \mathbf{x}) + \sigma^2 \mathbb{I} & K(\mathbf{x}, \mathbf{x}^\star) \\ K(\mathbf{x}^\star, \mathbf{x}) & K(\mathbf{x}^\star, \mathbf{x}^\star) \end{bmatrix} \right)$$

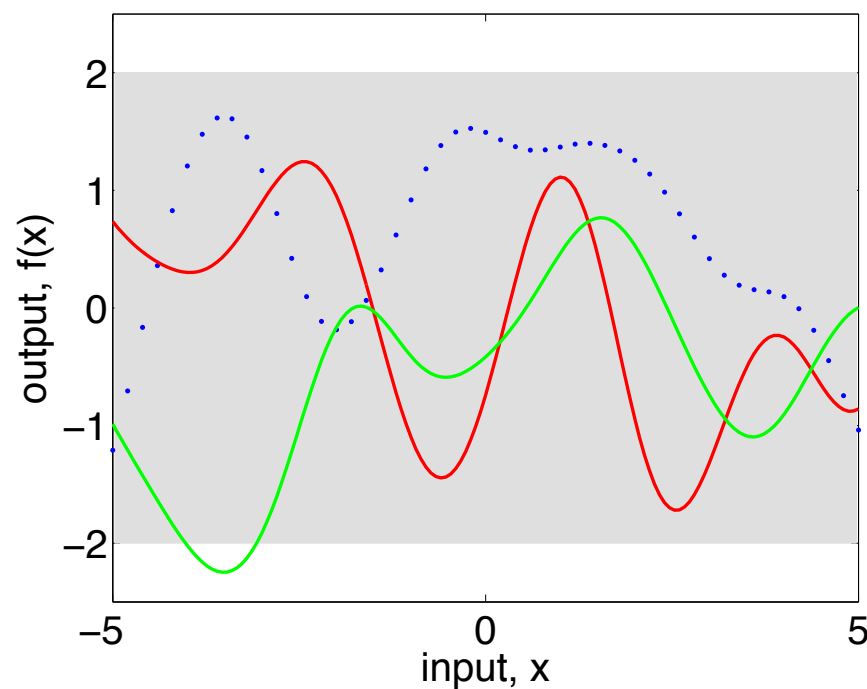
$$p(\mathbf{f}^\star | \mathbf{y}) \sim \mathcal{N} \left(K(\mathbf{x}^\star, \mathbf{x}) [K(\mathbf{x}, \mathbf{x}) + \sigma^2 \mathbb{I}]^{-1} \mathbf{y}, K(\mathbf{x}^\star, \mathbf{x}^\star) - K(\mathbf{x}^\star, \mathbf{x}) [K(\mathbf{x}, \mathbf{x}) + \sigma^2 \mathbb{I}]^{-1} K(\mathbf{x}, \mathbf{x}^\star) \right)$$

$$f(\mathbf{x}^\star) \sim \mathcal{N}(\mathbf{K}(K + \sigma^2 \mathbb{I})^{-1} \mathbf{y}, K(\mathbf{x}, \mathbf{x}) - \mathbf{K}^T (K + \sigma^2 \mathbb{I})^{-1} \mathbf{K})$$

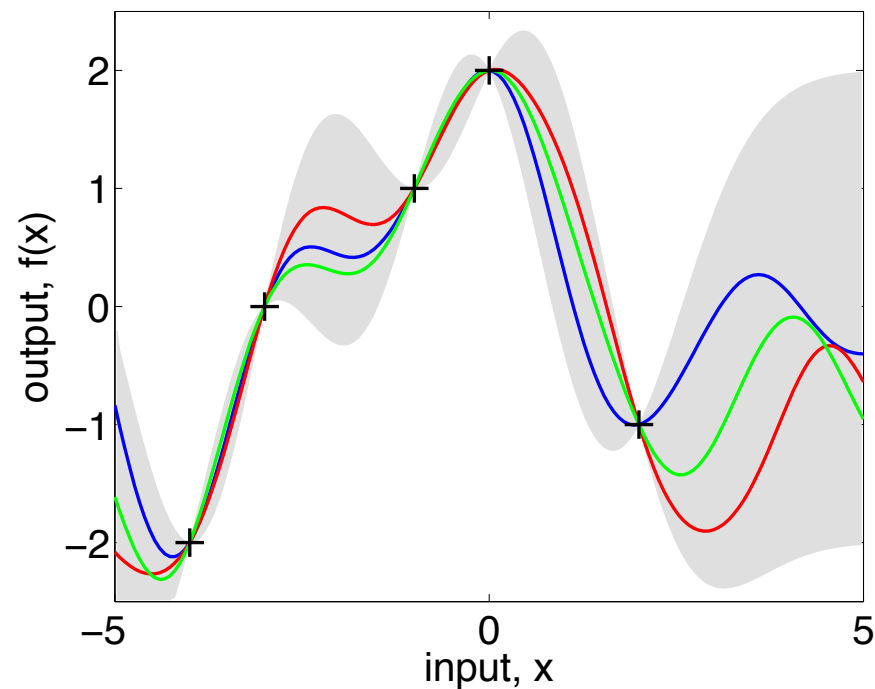
$$\mathbf{K} = (K(\mathbf{x}^\star, \mathbf{x})) = (k(\mathbf{x}^\star, x_1), \dots, k(\mathbf{x}^\star, x_N)).$$

Example

$$\mu(x) \equiv 0 \quad k(x, x') = \exp\left[-\frac{1}{2}|x - x'|^2\right] \quad \text{prior}$$



(a), prior



(b), posterior

Gaussian process, with mean

$$\mathbb{E}[f(\mathbf{x})|\mathbf{y}] = K(\mathbf{x}, X)(K + \sigma^2 I)^{-1}\mathbf{y}$$

and covariance

$$k(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') - K(\mathbf{x}, X)(K + \sigma^2 I)^{-1}K(X, \mathbf{x}')$$

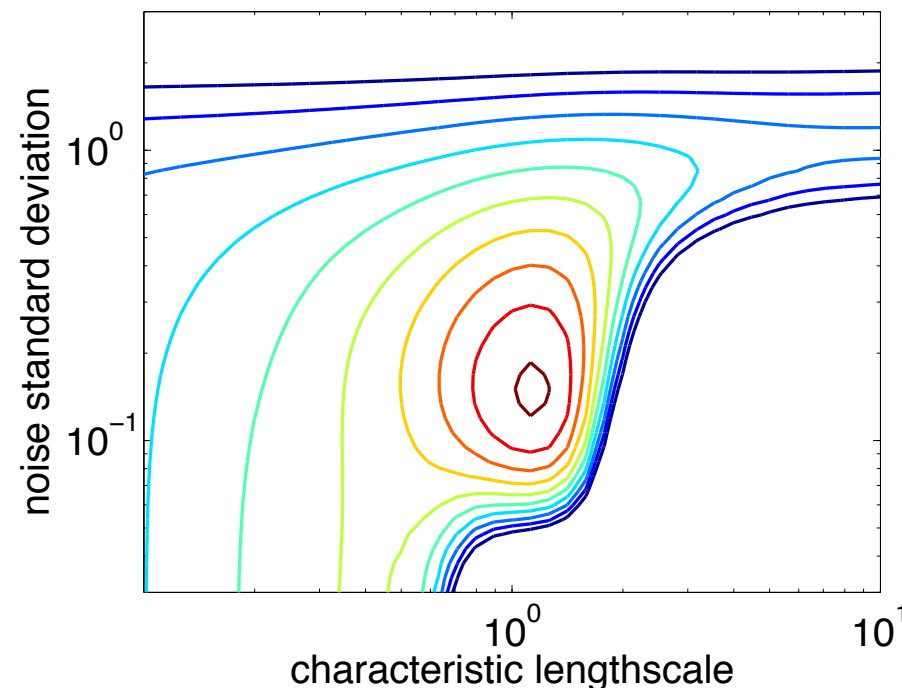
posterior

Hyper-parameter optimisation

Hyperparameters can be optimised by maximising the log marginal likelihood

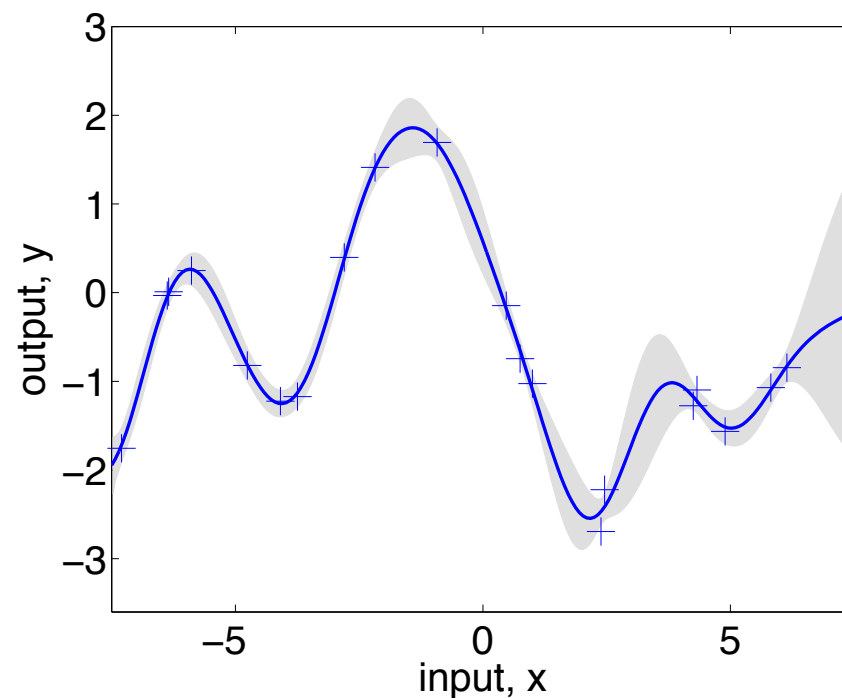
$$\mathcal{L} = -\frac{1}{2}\mathbf{y}^T (K + \sigma^2 \mathbb{I})^{-1} \mathbf{y} - \frac{1}{2} \log |(K + \sigma^2 \mathbb{I})| - \frac{N}{2} \log 2\pi$$

- ▶ $-\frac{1}{2}\mathbf{y}^T (K + \sigma^2 \mathbb{I})^{-1} \mathbf{y}$ which is the data fit;
- ▶ $-\frac{1}{2} \log |(K + \sigma^2 \mathbb{I})|$ which is a complexity penalty term;
- ▶ $-\frac{N}{2} \log 2\pi$ which is a constant term.

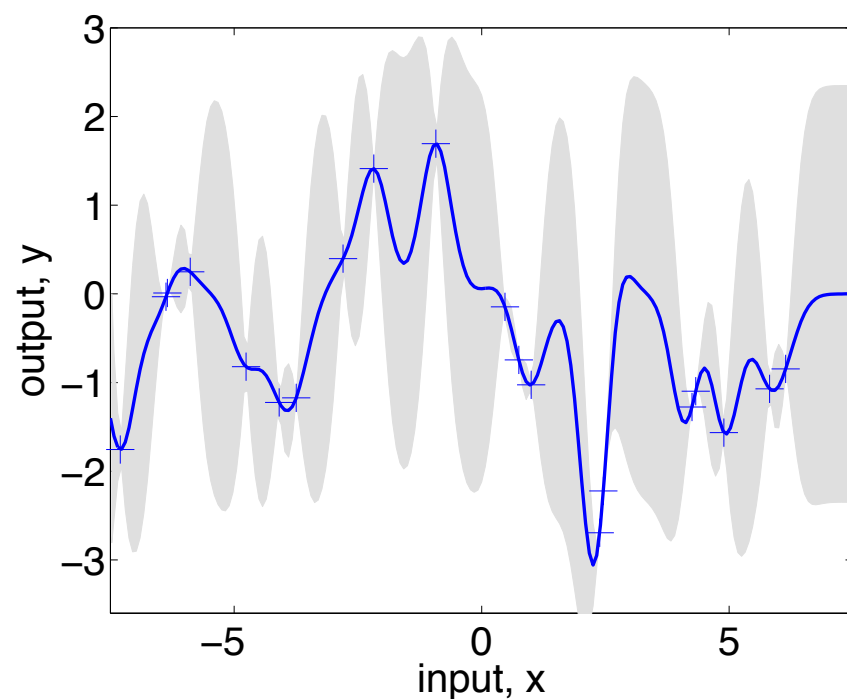


Dependence on hyperparameters

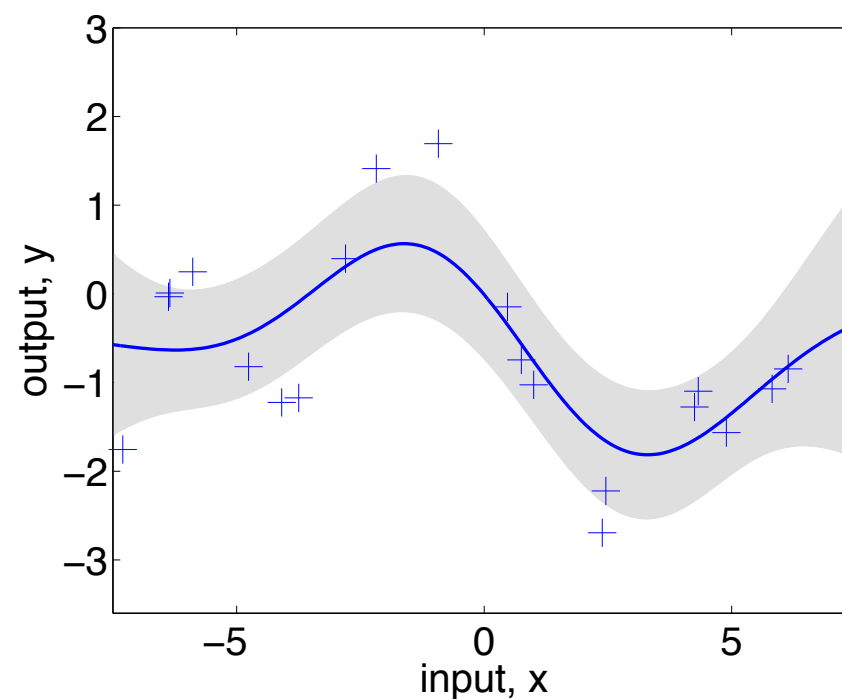
Gaussian kernel
lengthscale



(a), $\ell = 1$



(b), $\ell = 0.3$



(c), $\ell = 3$

